Retrieval-Augmented Event Relation Extraction: Efficient Causal and Temporal Relation Extraction via Dense Retrieval and Batch LLM Reasoning

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1 Motivation

Understanding how events relate to one another in time and causality is fundamental to natural language understanding. Event Relation Extraction (ERE) — identifying temporal and causal relations between events across documents — underpins applications including knowledge graph construction, question answering, and event prediction. MAVEN-ERE [1] is the largest unified ERE benchmark, covering over 1.2 million temporal relations and 57,000 causal relations across 4,480 English Wikipedia documents naturally rich in wars, elections, and political crises. Its large scale and unified annotation make it the most suitable benchmark for evaluating ERE systems where efficiency differences are meaningful and measurable.

Despite its importance, ERE at document scale remains unsolved. Standard approaches require pairwise comparison of all event mentions, resulting in quadratic complexity that scales poorly as documents grow longer. Efficient ERE [2] proposes a retrieve-then-classify pipeline improving F1 from 17.6 to 32.8 on all non-negative event pairs on MAVEN-ERE, but its retriever is fine-tuned only for binary relatedness rather than relation-type-specific signals, its cross-encoder produces no reasoning, and any pair dropped by the retriever is permanently lost. LLMERE [3] reformulates ERE as question answering, reducing model calls from $O(n^2)$ to $O(n)$ and achieving temporal F1 of 54.7 and causal F1 of 36.0 on MAVEN-ERE, but feeds entire unfiltered document partitions to the LLM, exposing it to noise that degrades precision and contributes to hallucination — confirmed on causal ERE by KnowQA [4]. Taco-ERE [5] addresses information overload through compression and clustering but without retrieval, while MAQInstruct [6] reduces inference samples by reframing queries around relation types. No existing work combines relation-type-aware dense retrieval with partitioned batch LLM reasoning and rationale generation in a unified pipeline for causal and temporal ERE. This gap motivates the present work.

2 Research Questions and Research Gap

RQ1: Can a relation-type-aware dense retriever effectively replace exhaustive pairwise enumeration for causal and temporal event relation candidate selection on MAVEN-ERE without significant loss in recall?

Efficient ERE shows retrieval improves overall F1 but uses a retriever fine-tuned only for binary relatedness rather than for causal and temporal relation signals specifically. The pairwise RoBERTa baseline achieves temporal recall of 53.9 and causal recall of 27.2 on MAVEN-ERE [1]. Whether a retriever fine-tuned on relation-type-specific signals can match this recall while substantially reducing the candidate space is an open question directly answerable through comparison on the MAVEN-ERE subset.

RQ2: Does presenting retrieved candidate event pairs in partitioned batches to an LLM maintain or improve relation classification accuracy and inference efficiency compared to full-document partitioning or single-pair classification?

LLMERE reports 14.4 seconds average inference time per document versus 90.3 seconds for the pairwise method, but feeds full unfiltered document partitions to the LLM [3]. The proposed approach applies LLMERE's partitioning strategy exclusively to pre-filtered retrieved candidates, so each partition contains only relevant events. This reduces noise per partition and total inference time while preserving LLM reasoning capability. Whether this improves over both Efficient ERE and LLMERE baselines on precision, recall, and inference time is directly answerable through comparison on MAVEN-ERE.

RQ3: Does requiring the LLM to generate a rationale for each classified relation pair improve precision on causal and temporal relations compared to classification-only approaches?

KnowQA [4] identifies hallucination — predicting relations where none exist — as a critical failure mode for LLM-based causal ERE. LLMERE introduces rationale generation and shows accuracy improvements overall but does not isolate the effect on precision specifically. Forcing the model to justify each prediction may act as a self-correction mechanism that reduces spurious predictions. Whether rationale generation measurably improves precision on causal and temporal relations is answerable through ablation within the same pipeline on the MAVEN-ERE subset.

3 Methodology

Proposed pipeline. The system consists of two stages. In the first stage, each event mention is wrapped with special markers following Efficient ERE [2] to direct encoder attention toward the target event within its sentence context. Each marked event is then encoded into a dense vector using a fine-tuned S-BERT bi-encoder. For each query event, FAISS nearest-neighbor search retrieves the top-k most relevant candidate events from the document efficiently without exhaustive pairwise comparison.

In the second stage, following LLMERE's partitioning strategy [3], the retrieved candidates for each query event are split into fixed-size groups controlled by a threshold n. Each group is presented jointly to the LLM in a single structured prompt. The prompt includes the source document context, the query event marked with a special identifier, the candidate events marked with sequential identifiers, and an instruction asking the model to output for each candidate pair the relation type — selecting from the causal and temporal

relation types defined in MAVEN-ERE — and a natural language rationale justifying the prediction. Results across partitions are merged to produce the final predictions for each query event.

Retriever training. The S-BERT bi-encoder is fine-tuned on MAVEN-ERE using a contrastive objective. Positive training pairs are event pairs annotated with a causal or temporal relation. Hard negatives are event pairs from the same document with no annotated relation, selected because they are most likely to confuse a similarity-based retriever. This contrastive training pulls positive pairs closer in embedding space and pushes hard negatives apart, producing a retriever sensitive to relation-type-specific signals rather than surface-level similarity alone.

Dataset. MAVEN-ERE [1] is the primary benchmark. Following Efficient ERE and prior work, evaluation is conducted on a fixed random-seed subset of the full dataset due to compute constraints, ensuring reproducibility. We follow both evaluation settings used by prior work: all event pairs, as used by LLMERE, and all non-negative event pairs, as used by Efficient ERE, to enable fair comparison across both baselines.

Evaluation measures. Primary metrics are precision, recall, and F1 per relation type (temporal, causal) following Wang et al. [1]. Secondary metrics are average inference time per document and number of LLM calls, compared against LLMERE's reported 14.4 seconds per document [3]. Baselines are: (1) pairwise RoBERTa from MAVEN-ERE (temporal F1=55.8, causal F1=30.6, temporal recall=53.9, causal recall=27.2 [1]); (2) Efficient ERE best configuration under all non-negative pairs (S-BERT + T5, F1=32.8 [2]); (3) LLMERE best configuration (Llama3-8B-base, temporal F1=54.7, causal F1=36.0 [3]). Two ablations are planned: batch classification versus single-pair classification to isolate the efficiency contribution of joint prompting, and rationale generation on versus off to isolate its effect on precision on causal relations specifically.

Tools and models. The retriever uses a fine-tuned S-BERT bi-encoder — the best performing retriever configuration in Efficient ERE on MAVEN-ERE — with FAISS for efficient nearest-neighbor search over event embeddings. The LLM for batch classification and rationale generation is Llama3-8B-base, the best performing model in LLMERE, enabling direct comparison under identical model conditions. Fine-tuning of the LLM uses LoRA via LLaMA-Factory following LLMERE's training setup, allowing parameter-efficient adaptation without updating all 8 billion parameters. Implementation uses Python, HuggingFace Transformers, FAISS, and LLaMA-Factory.

Git repository: https://git.fim.uni-passau.de/nourellil/ere-rag

4 Timeline and Implementation Plan

The project runs from May to September 2026 across three milestones aligned with the seminar deliverables.

Milestone 1 — Initial experiments (by 30th June 2026). The environment is configured, MAVEN-ERE is preprocessed and subsetted with a fixed random seed, the pairwise RoBERTa baseline is reproduced to validate the evaluation pipeline, the S-BERT bi-encoder is fine-tuned on the subset using a contrastive objective, FAISS indexing is built over the encoded event mentions, and initial end-to-end experiments are run with the partitioned batch LLM pipeline — all code and preliminary results committed to the Git repository. These initial results will form the basis of the seminar presentation in late June or early July 2026.

Milestone 2 — Refined experiments (by 1st September 2026). Both planned ablations are completed — batch versus single-pair classification to isolate the efficiency contribution of joint prompting, and rationale generation on versus off to isolate its effect on precision on causal relations — and all baseline comparisons against Efficient ERE and LLMERE are finalized and committed to the repository.

Milestone 3 — Wrap-up (by 10th September 2026). Results are fully analysed, conclusions and limitations are documented, and the final report is written and submitted by **10th September 2026**.

5 Limitations

Evaluation is conducted on a subset of MAVEN-ERE due to compute constraints, which may affect the generalizability of efficiency measurements to the full-scale setting. The retrieval stage introduces a hard recall ceiling — any true relation pair not retrieved in stage one cannot be recovered downstream — meaning the system trades some recall for efficiency, and the acceptable threshold of that tradeoff is an empirical question. Finally, Llama3-8B may produce inconsistent structured outputs in the batch prompt, requiring careful prompt engineering that could introduce brittleness in the evaluation pipeline.

AI Use Disclosure

Claude (Anthropic) was used to assist with the drafting and editing of this document and to explain technical concepts encountered during the literature review. All research direction decisions, paper selection, result extraction, and critical evaluation were conducted by the author (Hafsa NOUR ELLIL).

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